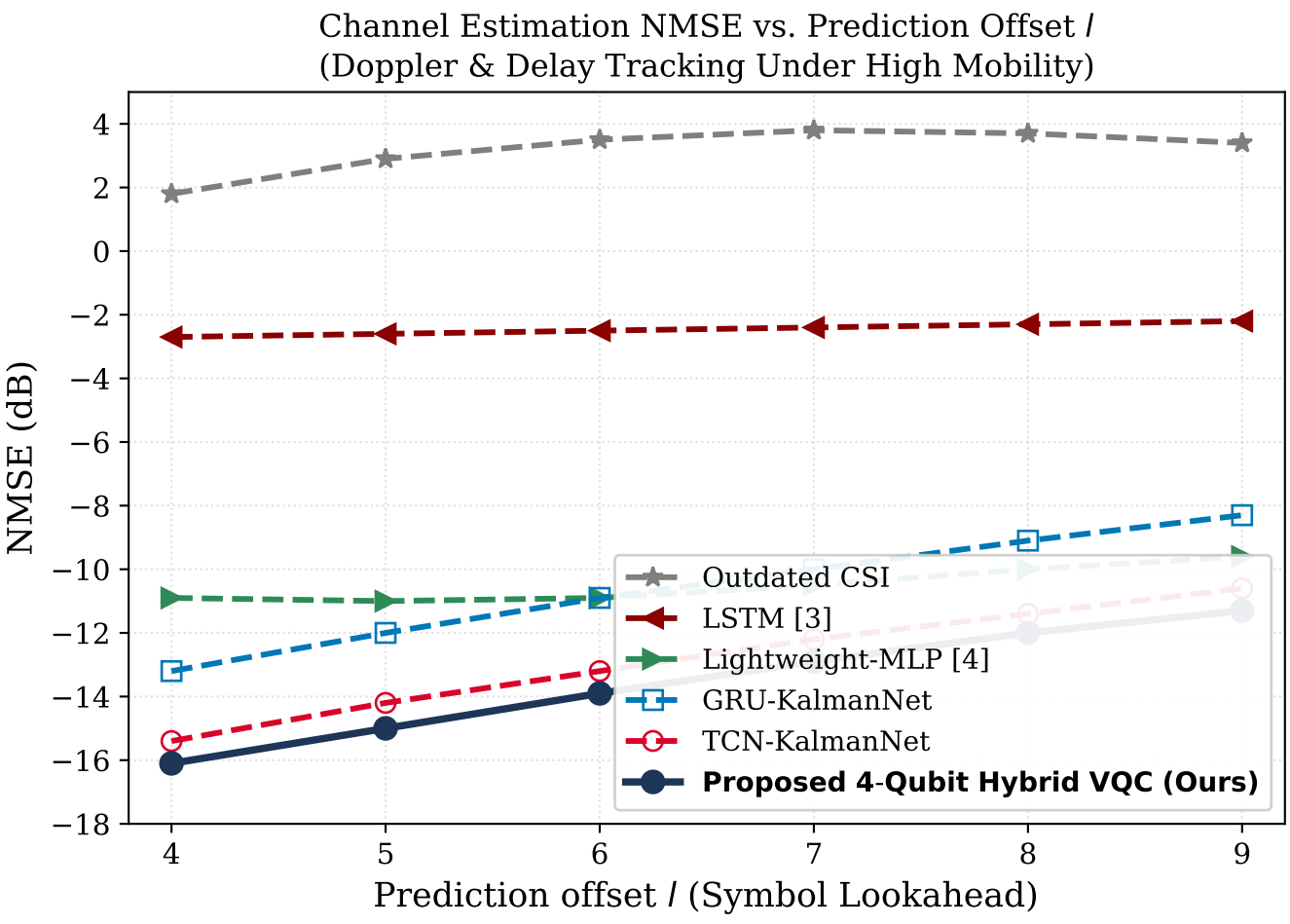
Channel Estimation NMSE vs. Prediction Offset /
(Doppler & Delay Tracking Under High Mobility)
NMSE (dB)
4
2
0
−2
−4
−6
−8
−10
−12
−14
−16
−18
4
5
6
7
8
9
Prediction offset / (Symbol Lookahead)
Outdated CSI
LSTM [3]
Lightweight-MLP [4]
GRU-KalmanNet
TCN-KalmanNet
Proposed 4-Qubit Hybrid VQC (Ours)